INCH-POUND

MIL-S-13261G (AR)
AMENDMENT 7
9 February 2001
SUPERSEDING
AMENDMENT 6
29 March 1991

MILITARY SPECIFICATION

SIGNAL, ILLUMINATION, GROUND, GREEN STAR CLUSTER M125A1, RED STAR CLUSTER M158, AND WHITE STAR CLUSTER M159, PARTS AND LOADING, ASSEMBLING AND PACKING

This amendment forms a part of MIL-S-13261G (AR), dated 10 January 1972, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Department and Agencies of the Department of Defense.

PAGE 1

* 2.1:

Specifications, Military. Delete "MIL-C-10464 Cans, Hermetic . . . Type" in its entirety.

Add: "MIL-C-70759 - Container, Ammunition, Plastic, PA142, for Pyrotechnic Hand Held Signal Series"

Under Standards, Federal. Delete "FED-STD-151 - Metals, Test Methods".

PAGE 2

2.1, Under Standards, Add:

"COMMERCIAL
ASTM E 8 – Tension Testing of Metallic Materials"

* 2.1, Drawings, delete "7548414" and "7548415" in their entirety and substitute the following:

"12900001 – Packing and Marking for Container, PA 142 for Signal, Ground
12900009 – Packing and Marking for Container, Metal M548, for Signals, Ground"

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:18Z
Check the source to verify that this is the current version before use.

* 3.4.1, Delete in its entirety and substitute:

"3.4.1 Premature burst. The assembly shall not burst within one hundred (100) feet from the launcher."

* Add new paragraph 3.4.1.1 and 3.4.1.2 as follows:

"3.4.1.1 Premature ejection. The illuminant assembly shall not eject at an attitude of less than one hundred (100) feet from the ground within a horizontal distance of two hundred and fifty (250) feet from the launcher, or at an altitude of less than fifty (50) feet from the ground at a horizontal distance in excess of two hundred and fifty (250) feet from the launcher.

3.4.1.2 Launch Delay. The signal must completely exit the rocket barrel within two (2) seconds of the striking of the primer."

PAGE 3

3.3.5: Cold temperature: Delete "3.3.5" and substitute "3.4.5".

3.4.2: Delete "Burning time of illuminant charges." And substitute "Burning time of illuminant charges."

3.4.3: Hot temperature:

Line 1, delete "(18 to 24°C or 65 to 75°F)" and substitute "(68 to 74°C or 155 to 165°F)".

* Delete "the requirements of 3.4.1" and substitute "the requirements of 3.4.1, 3.4.1.1 and 3.4.1.2."

3.4.4: Ambient temperature:

Line 1, delete "(70 plus or minus 5 degrees F)" and substitute "(18 to 24°C or 65 to 75°F)".

Change 4th subparagraph lettered "c" to subparagraph "d".

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:18Z
Check the source to verify that this is the current version before use.

* Delete “a. The requirements of 3.4.1 and 3.4.2” and substitute “a. The requirements of 3.4.1, and 3.4.1.1, 3.4.1.2 and 3.4.2”.

* 3.4.5: Cold temperature. Delete “a. The requirements of 3.4.1 and 3.4.2” and substitute “a. The requirements of 3.4.1, 3.4.1.1, 3.4.1.2 and 3.4.2”.

* 3.5: Leakage. Delete “metal container” and substitute “plastic container”.

* 3.6: Container tear strip. Delete in its entirety.

* 3.9: Add new paragraph:

“3.9 Airtightness. The metal container shall withstand, without leakage, an air pressure differential of three (3) pounds per square inch (psi) when tested as specified in 4.5.13.”

PAGE 5

Table I, Under Signals, Assemblies Delete: “Functioning, cold, hand fired, 16, 100%, 3.4.5, 4.5.2.3.1/hand”

* Table I:

Delete “(Dwg.7548414)” and substitute “(dwg. 12900001)”.

Delete “Tear Strip Test (from the Leakage test)” in its entirety.

Add: “Container, Metal, M548 (Dwg. 12900009)
Examination for defects, 4 metal containers, 3.2, 4.4.2.24”

Add Note 3 as follows:

“Note 3. If any if the four (4) M548 containers are found defective. It should be replaced with an acceptable container prior to functioning tests”.

Under containers (dwg. 12900001) delete “Leakage test 96” and substitute Leakage test 80.”

PAGE 12

4.4.2.6: Note (a): Delete “200 samples” and substitute “20 samples”.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:18Z
Check the source to verify that this is the current version before use.

PAGE 13

4.4.2.7: Note (b): Delete the first sentence, “200 samples ...” and substitute “20 (twenty) samples/lot or batch – same sample for both tests.”

PAGE 18

4.4.2.12: Tail Assembly: Delete in its entirety and substitute included page.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:18Z
Check the source to verify that this is the current version before use.

Quality Conformance Inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Tail Assembly				8797947
					NEXT HIGHER ASSEMBLY 8797920
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical					
1	Tail fin missing, cracked or inverted		100%	3.2	Visual
Major					
101	Length, max		0.40%	3.2	Gage
102	Location of tail vanes with diameter and each other, including maximum vane extension from centerline		0.40%	3.2	Gage
103	Inside diameter of upper tail ring, min		0.40%	3.2	Gage
104	Perpendicularity of ends with diameter		0.40%	3.2	Gage
105	Position of rib inside diameter with upper tail ring diameter		0.40%	3.2	Gage
106	Weld missing		0.40%	3.2	Gage
107	Weld flash excessive to extent that function will be impaired		0.40%	3.2	Visual
108	Weld peel test	(a)	---	3.2	4.4.11
109	Mechanical properties test (See Dwg. 8797951)	(b)	---	3.2	4.5.8
Minor					
201	Clearance between lower tail ring and tail vane		0.65%	3.2	Gage
203	Poor workmanship		1.00%	3.8	Visual

NOTES: 32 samples/lot, 1 upper tail ring weld and 1 lower tail ring weld per assembly (64 weld total); accept 0, reject 1. (b) Five (5) tail vanes/test samples shall be selected from each heat treated batch for mechanical properties testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:18Z
Check the source to verify that this is the current version before use.

PAGE 20

4.4.2.14: Tube, casing: Delete defects in their entirety and substitute the following:

"Critical				
1	Hydrostatic pressure test	100% (Note 1)	3.2	4.5.6
2	Presence of cracks	100%	3.3	Visual
Major				
101	Perpendicularity of ends	0.40%	3.2	Gage
102	Total length, max.	0.40%	3.2	Gage
103	Inside diameter at ends, max.	0.40%	3.2	Gage
Minor				
201	Poor workmanship	1.00%	3.8	Visual

Note 1 - Tubing stock used to manufacture the casing tube shall be 100% hydrostatically tested as specified in Para. 4.5.6. All remaining classified defects shall be checked on the finished item."

PAGE 24

* 4.4.2.18:

Add critical defect: "3 Primer above flush 100% 3.2 gage".

Add major defect " 104 Depth of primer incorrect, .40%, 3.2, gage"

PAGE 28

* 4.4.2.22: Delete "(Dwg. 7548414)" and substitute "(dwg. 12900001)". Delete "(Dwg. 7548415)" and substitute "(dwg. 12900009)".

PAGE 29

* 4.4.2.23:

Delete dwg. "7548414" and "7548415" and substitute "121900001" and "12900009".

Critical 1: Delete in its entirety.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:18Z
Check the source to verify that this is the current version before use.

Critical 2: Add: "Cover" before "Embossing . . .".

Majors 101 and 104: Delete in their entirety.

Add Major 105: "105 Label missing or improperly applied, -, 0.40%, 3.2, Visual"

Minor 203: Delete in its entirety.

PAGE 30

* Major Defect 102, delete in its entirety and substitute the following:

"102 Air tightness test (after sealing) Note 1 3.9 4.5.13"

* Add Note 1 as follows:

"Note 1. One container out of each 20 containers packed out/sealed shall be tested. If the container fails the test, it should be removed from the lot and all containers back to the last successful test should be leak tested."

* 4.4.2.24: Delete in its entirety and substitute the included page 30.

4.4.2.25: Add new Classification of defects for Signal Body (Dwg. 8797963) as included page 30a.

* 4.4.2.26: Add new paragraph 4.4.2.26 as included page 30b.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:18Z
Check the source to verify that this is the current version before use.

Quality Conformance Inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH 4.4.2.24	TITLE Packing and Markings for Container, Metal M548 for Signals, Ground		SHEET 1 OF 1		DRAWING NUMBER 12900009 NEXT HIGHER ASSEMBLY ----
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined				
Major					
101	PA142 Container quantity incorrect, or improperly inserted		0.40%	3.2	Visual
102	Air tightness test (after sealing)	NOTE 1	0-1	3.9	4.5.13
103	Metal container damaged (dents, splits, etc.)		0.40%	3.2	Visual
Minor					
201	Contents loose		0.65%	3.2	Visual
202	Markings Misleading or unidentifiable		0.65%	3.2	Visual
203	Metallic seal missing, unsealed or improperly Engaged		0.65%	3.2	Visual
204	Poor workmanship		1.00%	3.8	Visual

NOTES: Note 1. One container out of each 20 containers packed out/sealed shall be tested. If the container fails the test, it should be removed from the lot and all containers back to the last successful test should be leak tested.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:18Z
Check the source to verify that this is the current version before use.

Quality Conformance Inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH 4.4.2.25	TITLE Signal Body		SHEET 1 OF 1		DRAWING NUMBER 8797963 NEXT HIGHER ASSEMBLY 8797920
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined				
Major					
101	Small inside diameter		0.40%	3.2	Gage
Minor					
201	Large outside diameter, max		0.65%	3.2	Gage
202	Large inside diameter, min.		0.65%	3.2	Gage
203	Overall length		0.65%	3.2	Gage
204	Poor workmanship		1.0%	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:18Z
Check the source to verify that this is the current version before use.

Quality Conformance Inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH 4.4.2.26	TITLE Protector Assembly		SHEET 1 OF 1		DRAWING NUMBER 8797941 NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical Major 101 Minor	None defined Outside diameter, max None defined		100%	3.2 (See Dwg. 8797941)	Gage
NOTES:					

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:18Z
Check the source to verify that this is the current version before use.

PAGE 31

4.4.3.1: TABLE II. Under “Materials” heading, Delete the word “Fiberboard” and substitute “Hardboard”.

PAGE 32

4.4.3.2: Table III, eleven lines below “Defect Classification,”: Delete the reference, (“See 3.e.4d)” and substitute “(See 3.4.4d)”.

* Table III:

Add the following characteristics:

“Premature ejection (See 3.4.1.1)	Special (see 6.8)
Launch Delay (See 3.4.1.2)	Major”

Delete “Assembly bursts at a distance of more than one hundred (100) feet from launcher (see 3.4.1) Major” and substitute “Assembly bursts at a distance at a distance of more than one hundred (100) feet from the launcher but less than the distance specified in para. 3.4.1.1 Special”

4.4.3.2.1.1:

Delete “one hundred (100) assemblies” and substitute “fifty (50) assemblies.”

* Delete “a critical defect” and substitute “a critical or special defect,”.

4.4.3.2.1.2:

Delete “twenty-five (25) assemblies” and substitute “twenty (20) assemblies.”

* Delete “a critical defect” and substitute “a critical or special defect,”.

4.4.3.2.2.1:

Delete “one hundred and twenty-five (125) assemblies” and substitute “one hundred (100) assemblies.”

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:18Z
Check the source to verify that this is the current version before use.

Delete “or if eight (8) or more” and substitute “or if six (6) or more.”

* Delete “a critical defect occurs,” and substitute “a critical or special defect occurs,”.

PAGE 33

* 4.4.3.2.2.2: Delete “a critical defect occurs,” and substitute “a critical or special defect occurs,”.

4.4.3.2.3.1: Delete entire paragraph and substitute the following:

“4.4.3.2.3.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have been found to be acceptable, fifty (50) assemblies shall be selected from each lot for test. The lot shall be rejected if, during the test, a critical defect occurs, the average angle from the vertical exceeds the requirement, of if (4) or more assemblies exhibit any of the remaining major defects listed in Table III. (See note after the following paragraph).”

* 4.4.3.2.3.1, Delete “a critical defect occurs,” and substitute “a critical or special defect occurs,”.

4.4.3.2.3.2: Delete entire paragraph and substitute the following:

“4.4.3.2.3.2 After three (3) consecutive lots have compiled with the criteria of 4.4.3.2.3.1, thirty-two (32) assemblies shall be selected from each lot for test. The lot shall be rejected, if during the test, a critical defect occurs, the average angle from the vertical exceeds the requirement or if three (3) or more assemblies exhibit any of the remaining major defects listed in Table III. (see note)

Note: A star which ignites and burns normally but subsequently fails to meet minimum burning time requirement due to ground impact, will not be classified defective. Firing records will be annotated accordingly.”

* 4.4.3.2.3.2, Delete “a critical defect occurs,” and substitute “a critical or special defect occurs,”.

4.4.3.2.4: Cold temperature, hand fired: Delete in its entirety.

PAGE 34

4.4.3.3: Propellant ballistic test. Second sentence, delete “ as classified in Table III of 4.4.3.2” and substitute “(See para. 3.4.1),”.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:18Z
Check the source to verify that this is the current version before use.

PAGE 36

4.5.2.1:

* Line four, delete “metal” and substitute “plastic”.

Note. Delete “60 seconds” and substitute “three minutes”.

4.5.2.2: Note. Delete “60 seconds” and substitute “three minutes”.

PAGE 37

4.5.2.3.1:

* Line four, delete “metal” and substitute “plastic”.

Note. Delete “60 seconds” and substitute “three minutes”.

4.5.2.3.2: Hand fired: Delete in its entirety.

* 4.5.3: Leakage: Line 1, delete “The packed sealed container” and substitute “The packed plastic container (dwg. 12900001)”

* 4.5.4: Container tear strip test. Delete in its entirety.

PAGE 38

4.5.6: Delete in its entirety and substitute the following:

“4.5.6 Hydrostatic test of tube, casing. In lieu of testing each individual casing tube, the tubing stock, which is used to manufacture the casing tube, shall be 100% tested in a hydrostatic test device (see 4.4.4). The minimum pressure shall be applied for not less than 5 seconds. The tubing shall be observed for evidence of failure to comply with the requirements. Any full length of tubing that bursts, cracks, or shows evidence of distortion shall be classified as defective and removed from production.”

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:18Z
Check the source to verify that this is the current version before use.

4.5.8: Hardness of tall vane: Delete in its entirety and substitute:

"4.5.8 Mechanical properties of tail vane: The tail vane/test samples shall be tested for mechanical properties in accordance with ASTM E8. If any tail vane/test sample fills to comply with the criteria specified on the applicable drawing, the heat treated batch represented by the sample shall be rejected."

4.5.9: Push test of firing cap and spring assembly. Delete the second sentence, "If any sample…" and substitute " If any sample fails to comply with the applicable requirement, it shall be classed defective."

PAGE 39

* Add the following new paragraphs:

"4.5.13 Airtightness. The metal container shall be conditioned to ambient temperature and pressure prior to testing. The container shall be tested in accordance with 4.5.13.1 or with permission of the procurement agency with 4.5.13.2. A wetting agent may be used to minimize air bubbles clinging to the exterior surface. A stream or recurring succession of bubbles from any surface, seam or junction shall be evidence for a defective container. Any container that fails to comply with the requirement shall be classified defective and removed from the lot.

4.5.13.1 Vacuum method. The container shall be tested for leakage by immersing the closed container in the inverted position under water in a vacuum vessel and lower the pressure in the vessel to three pounds per square inch minimum below ambient pressure. Observation for leakage of air from the container interior shall be made for a minimum of three seconds after reduction of pressure.

4.5.13.2 Hot water method. The container shall be tested for leakage by immersing the closed container in the inverted position to a depth of one inch below the surface of the water. The temperature of the water and the length of time of immersion shall be that which will assure an increase in pressure to a minimum of 3 psi above ambient pressure. Observation for leakage of air from the container interior shall be made during the period of pressure buildup and for 30 seconds after the 3 psi pressure differential has been reached".

* 5.1.1: Delete "7548414" and substitute "12900001".

* 5.2.1: Delete "7548415" and substitute "12900009".

* 5.3: Delete "7548414 and 7548415" and substitute "12900001 and 12900009."

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:18Z
Check the source to verify that this is the current version before use.

6.4: Hand firing: Delete in its entirety.

6.7: Delete in it entirety and substitute:

"6.7 Proving ground functioning test summary production lots.

Firing Method & Conditioning Temp. Requirement	First Article	First Three Consecutive Acceptable	Remaining Acceptable
Fixture: 68° to 74°C See 3.4.3 (155° to 165°F)	32	50	20
18° to 24°C See 3.4.4 (65° to 75°F)	32	100	50
-57° to -51°C See 3.4.5" (-70° to -60°F)"	16	50	32

* Add new paragraph 6.8 as follows:

"6.8 Definitions:

a. Special defect. A defect, other than critical that judgment and experience indicate may, depending upon the degree of variance from the design requirement:

1. Results in hazardous or unsafe condition for individuals using, maintaining or depending upon the product.

2. Prevent performance of the tactical function of a major end item.

b. Premature burst. A signal which ejects, ignites and burns at a distance less than that specified in the requirement paragraph (see 3.4.1).

c. Premature ejection. A signal in which the illuminant assembly ejects but fails to ignite (see 3.4.1.1)".

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:18Z
Check the source to verify that this is the current version before use.

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Army-AR

Preparing activity
Army-AR

(Project 1370-0030)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:18Z
Check the source to verify that this is the current version before use.